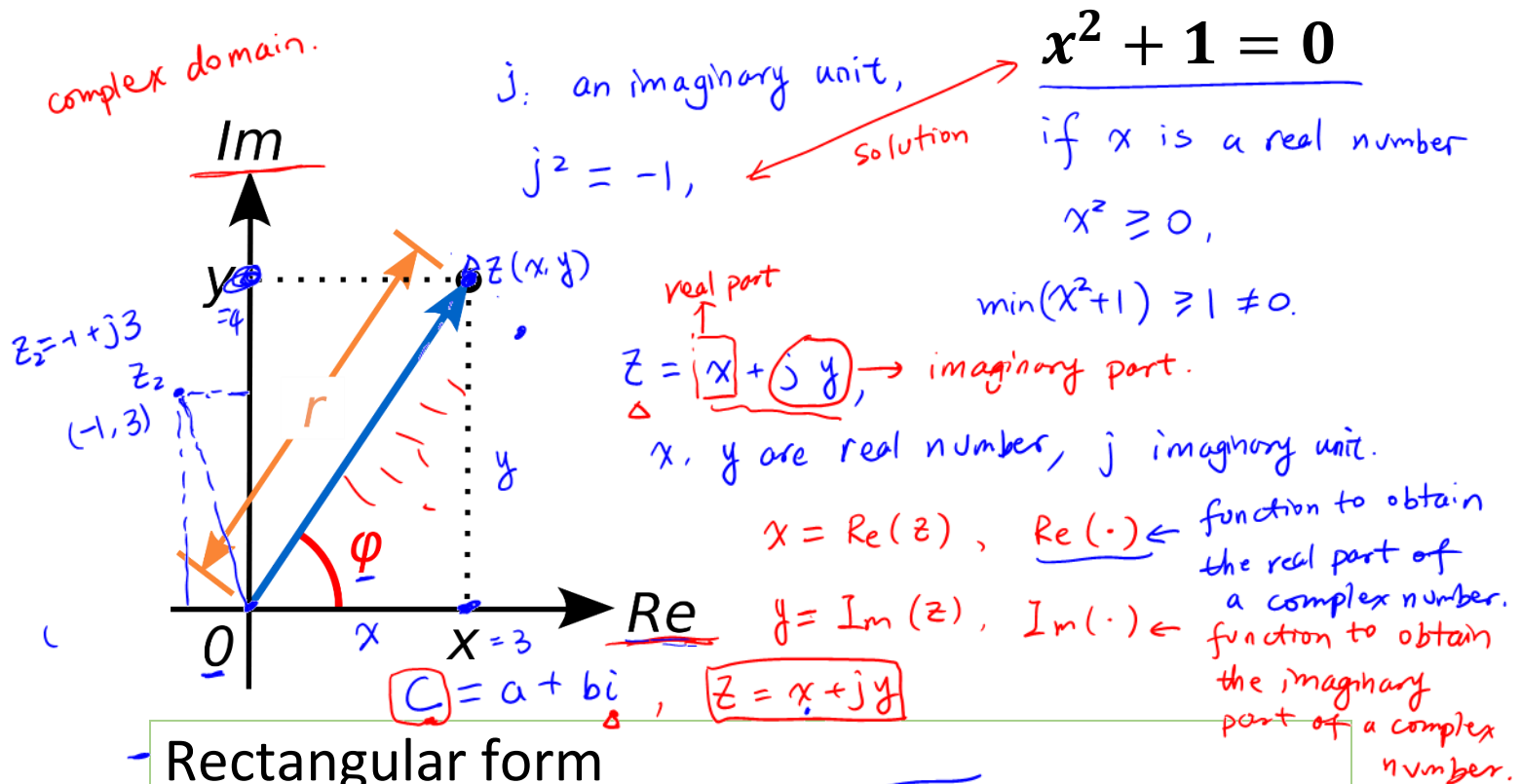


## L#27: Complex numbers



### - Rectangular form

$$z = x + jy, \quad z = 3 + j4, \quad r = \sqrt{x^2 + y^2}$$

- ① Given  $z = x + jy$ , obtain its polar form.
- $$\tan \varphi = \frac{y}{x}, \quad \varphi = \tan^{-1}\left(\frac{y}{x}\right).$$
- $$z = r \angle \varphi.$$

### - Polar form $z = r \angle \varphi$

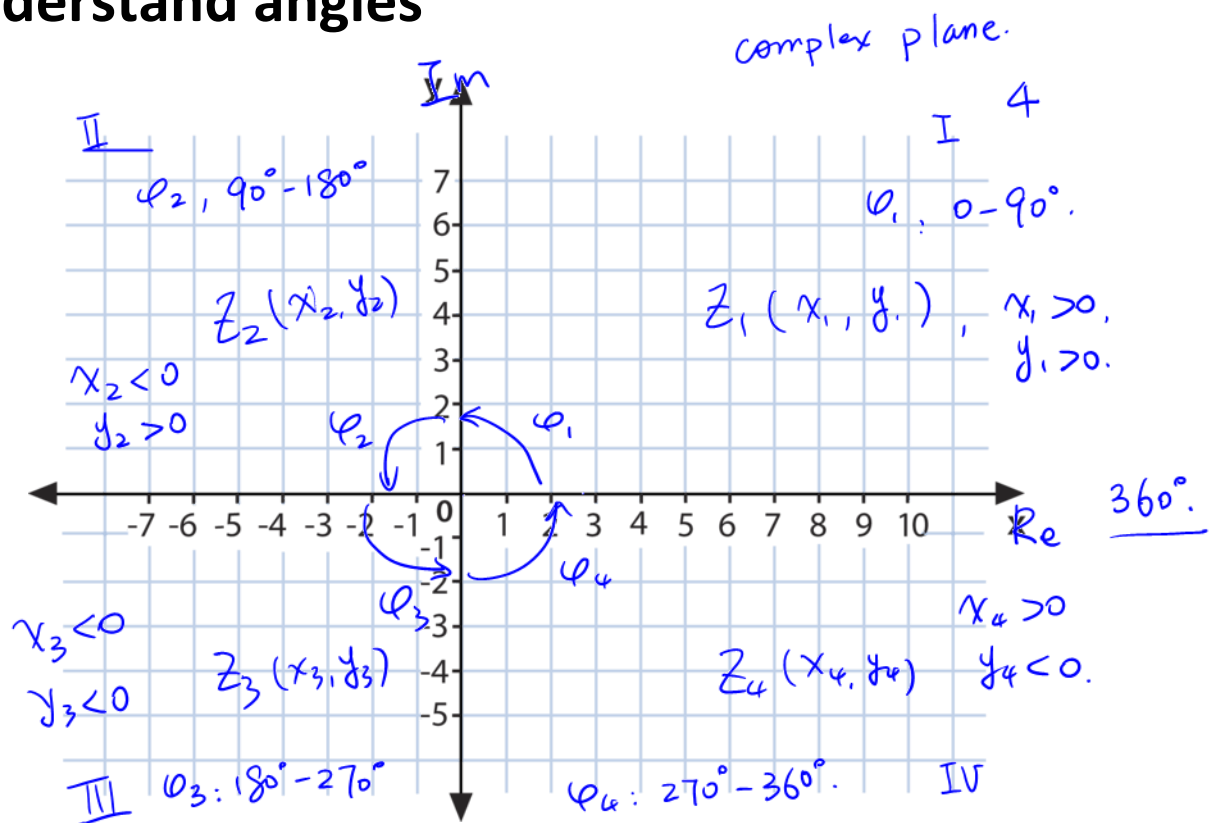
- ② Given  $z = r \angle \varphi$ , obtain rectangular form.

$$z = x + jy, \quad x = r \cos \varphi, \quad y = r \sin \varphi.$$

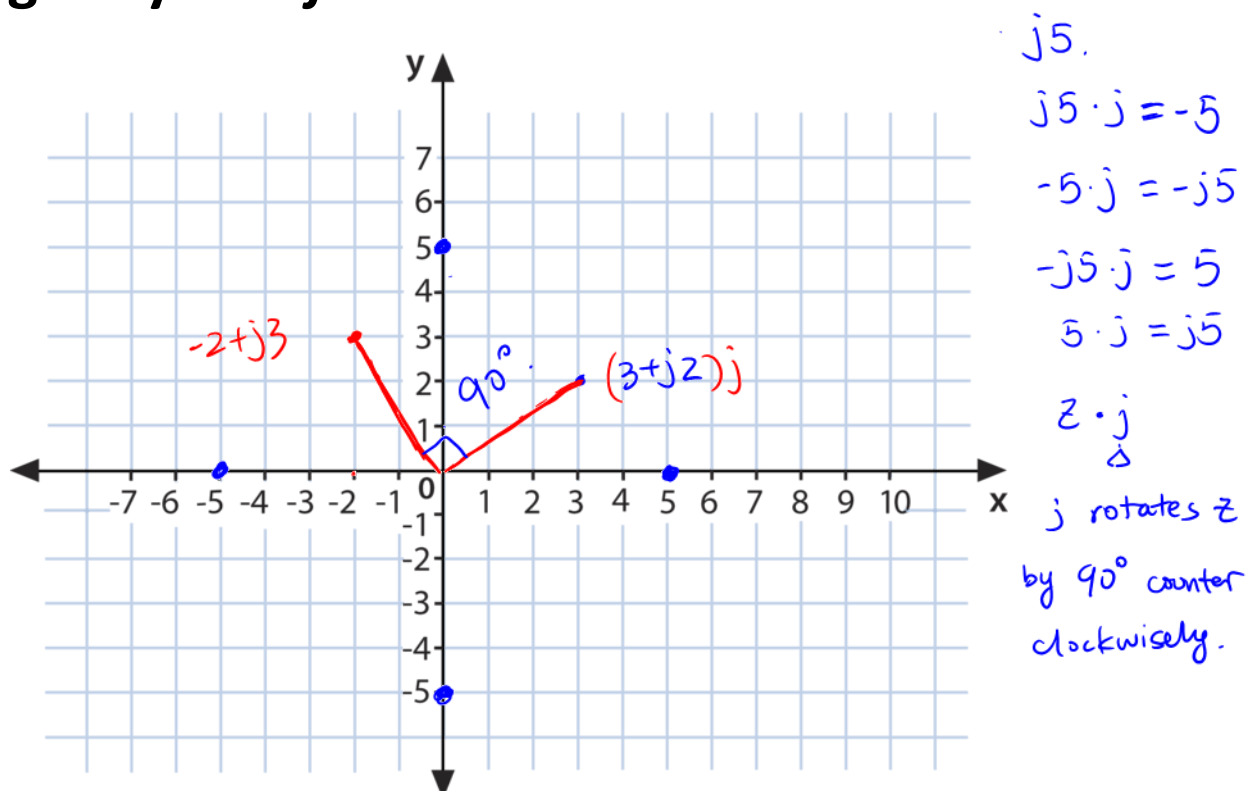
$$z = r \cos \varphi + jr \sin \varphi = r(\underbrace{\cos \varphi + j \sin \varphi}_{e^{j\varphi}}) = r e^{j\varphi}.$$

### - Exponential form

# Understand angles



## Imaginary unit $j$



## Conversion between forms

$$\underline{z_1} = 2 + j6 : \text{rect form.}$$

polar form and exp form.

$$z_1 = r \angle \theta.$$

$$r = \sqrt{2^2 + 6^2} = 6.32$$

$$\phi = \tan^{-1}\left(\frac{6}{2}\right) = \tan^{-1}(3) = \underline{71.57^\circ}$$

$$z_2 = 2e^{j30^\circ} \text{ exp. form.}$$

$$= 2 \angle 30^\circ$$

$$= 2 (\cos 30^\circ + j \sin 30^\circ)$$

$$= 1.73 + \boxed{j1} \rightarrow 1.$$

$$z_1 = 6.32 \angle 71.57^\circ$$

$$\underline{z_1 = 6.32 \angle 71.57^\circ}$$

$$z_1 = 6.32 e^{j71.57^\circ}$$

$$z_3 = 5 \angle 275^\circ, \text{ Polar form.}$$

$$= 5 e^{j275^\circ}$$

$$= 0.44 - j4.98.$$

$$z = \frac{a + bi}{\Delta} \rightarrow \text{imaginary unit.}$$

# Complex number arithmetic

## Addition and subtraction

$$z_1 = x_1 + jy_1 \quad z_2 = x_2 + jy_2$$

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

$$z_1 = r_1 \angle \phi_1, \quad z_2 = r_2 \angle \phi_2.$$

$$\underline{z_1 + z_2 \neq (r_1 + r_2) \angle (\phi_1 + \phi_2)} \quad \times$$

addition and subtraction  
not work with  
polar form for  $z$ .

$$\underline{z_1 = 2.5 + j4.6}$$

$$\underline{z_2 = 5 \angle 275^\circ}$$

$$= 0.44 - j4.98.$$

$$z_1 + z_2 =$$

# Multiplication

$$z_1 = x_1 + jy_1$$

Rectangular & Polar forms

$$z_2 = x_2 + jy_2$$

$$\begin{aligned} z_1 z_2 &= x_1 x_2 + jx_1 y_2 + jx_2 y_1 + j^2 y_1 y_2 \\ &= x_1 x_2 + j(x_1 y_2 + x_2 y_1) - y_1 y_2. \end{aligned}$$

$$z_1 = r_1 \angle \phi_1, \quad z_2 = r_2 \angle \phi_2,$$

$$z_1 z_2 = r_1 r_2 \angle (\phi_1 + \phi_2). \quad \checkmark$$

$$\frac{(x_1 + jy_1)}{z_1} \times \frac{r_1 \angle \phi_1}{z'_1}$$

can't calculate directly.

$$z_1 = 2.5 + j4.6$$

$$z_2 = 5 \angle 275^\circ$$

$$= 5.24 \angle 61.48^\circ.$$

$$z_1 z_2 = 26.18 \angle \underline{-23.52^\circ}$$

$$(275^\circ + 61.48^\circ) - 360^\circ.$$

# Divisions

$$z_1 = x_1 + jy_1$$

Rectangular & Polar forms

$$z_2 = x_2 + jy_2$$

$$\frac{z_1}{z_2} = \frac{x_1 + jy_1}{x_2 + jy_2} \cdot \frac{x_2 - jy_2}{x_2 - jy_2} = \frac{x_1 x_2 + j(x_2 y_1 - x_1 y_2) + y_1 y_2}{x_2^2 + y_2^2}$$

$$= \frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} + j \frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2}$$

Re im.

$$z_1 = r_1 \angle \theta_1$$

$$z_2 = r_2 \angle \theta_2$$

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle (\theta_1 - \theta_2)$$

simpler to calculate  
in polar form.

$$z_1 = 2.5 + j4.6$$

$$z_2 = 5 \angle 275^\circ$$

$$= 5.24 \angle 61.48^\circ$$

$$\frac{z_1}{z_2} = 1.05 \angle \underline{61.48 - 275 + 360^\circ}$$

# Root

$$\mathbf{z_1 = x_1 + jy_1}$$

Rectangular & Polar forms

$\times \rightarrow$  you get an error from the calculator.  
if direct root a rect. form of  $\mathbf{z}$ .  
only works for polar form.

$$\mathbf{z_1 = r_1 \angle \theta_1}$$

$$\sqrt{\mathbf{z_1}} = \sqrt{r_1} \angle \frac{\theta_1}{2}.$$

$$\sqrt[n]{\mathbf{z_1}} = \sqrt[n]{r_1} \angle \frac{\theta_1}{n}, \quad n \text{ is an integer.}$$

$$\mathbf{z_1 = 2.5 + j4.6}$$

$$\mathbf{z_2 = 5 \angle 275^\circ}$$

$$\sqrt{\mathbf{z_2}} = 2.24 \angle 137.5^\circ.$$

# Inverse

$$\mathbf{z_1 = x_1 + jy_1}$$

Rectangular & Polar forms

$$\frac{z_2}{z_1}, \text{ when } z_2 = 1.$$

$$\mathbf{z_1 = r_1 \angle \theta_1}$$

$$\text{inverse } z_1 \Rightarrow \frac{1}{z_1}.$$

$$\frac{1}{z_1} = \frac{1}{x_1 + jy_1} \cdot \frac{(x_1 - jy_1)}{(x_1 - jy_1)} = \frac{x_1 - jy_1}{x_1^2 + y_1^2}.$$

$$\frac{1}{z_1} = \frac{1}{r_1 \angle \theta_1} = \frac{1 \angle 0^\circ}{r_1 \angle \theta_1} = \frac{1}{r_1} \angle (0^\circ - \theta_1) = \frac{1}{r_1} \angle -\theta_1$$

$$\mathbf{z_1 = 2.5 + j4.6}$$

$$\mathbf{z_2 = 5 \angle 275^\circ}$$

$$\frac{1}{z_2} = \frac{1}{5} \angle -275^\circ$$

$$= 0.2 \angle -275^\circ$$

$$= 0.2 \angle 85^\circ.$$

$$\underline{\underline{-275^\circ + 360^\circ}}$$

$$\downarrow$$

$$-180^\circ \leq \phi \leq 180^\circ.$$



# Conjugate (\*)

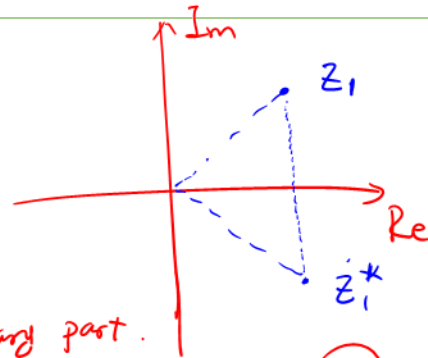
$$z_1 = x_1 + jy_1$$

Rectangular & Polar forms

①

$$z_1^* = x_1 - jy_1$$

change the sign of imaginary part.



③

$$z_1 = r_1 \angle \theta_1$$

②

$$z_1^* = r_1 \angle -\theta_1$$

change the sign of the angle.

$$z_1 = 2.5 + j4.6$$

$$z_1^* = 2.5 - j4.6$$

$$z_2 = 5 \angle 275^\circ$$

$$z_2^* = 5 \angle -275^\circ$$

**Example #1: answer in three forms.**

$$z = \frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5$$

$$= 10.983\angle 40.968^\circ$$

$$= 8.293 + j7.2$$

$$= 10.983 e^{j40.968^\circ}$$